

IBONG TIRIRIT (PIPE 9)

- Is the most common dryer used which consist of rotating cylinder inside which the materials flow while getting in cont act with hot gas
 - **Rotary Dryer**
- Is the ratio of the mass of water - vapor in air and mass of air if it is saturated is called
 - **Relative Humidity**
- The refrigerant use in steam jet cooling is
 - **Water**
- The total heat of air is a function of
 - **WB temperature**
- Boiling point of Freon - 12 at atmospheric pressure is
 - **21 degree Fahrenheit**
- Which of the following is NOT a type of water cooled condenser in refrigeration
 - **Double Shell**
- Component of absorption refrigeration system in which the solution is cooled by cooling water
 - **Absorber**
- Cascade refrigeration cycle is often used in industrial process where objects must be cooled to temperature below
 - **46 deg C**
- Type of refrigerant control designed to maintained a pressure difference while the compressor is operating
 - **Capillary tube**
- As a rule of thumb, for specified amount of compressed air, the power consumption of the compressor decreased by ____ for each 3C drop in the temperature inlet air to the compressor
 - **1%**
- Modern way of detecting air compressor leak is by using
 - **Acoustic Leak Detector**
- For foundation of stacks, the maximum pressure on the soil is equal to the pressure due to weight and the _____.
 - **Wind Movement**
- Foundation of bolts of specified size should be used and surrounded by a pipe sleeve with an inside diameter of at least...
 - **3 times the diameter of engine bolt**
- For multi stage compression of an ideal Brayton Cycle, the back work ratio will
 - **Decrease**
- Type of turbine that has a specific heat below 5
 - **Impulse turbine**
- A high discharge type of turbine
 - **Propeller turbine**
- Use to minimize the speed rise due a sudden load rejection
 - **Jet Deflector**
- Is the speed of a turbine when the head on the turbine is 1 meter
 - **Unit Speed**
- A fluid property which refers to the intermolecular attraction by which the separate particles of the fluid arc held together
 - **Cohesion**
- Which of the following is NOT the cause of black smoke in diesel?
 - **High Compression Pressure**
- Which of the following is NOT a method of starting a diesel engine?
 - **Using another generator**
- Two stroke engine performs ____ to complete one cycle.
 - **Compression and Power stroke**
- A type of geothermal plant used when there is a presence of brine extracted from underground
 - **Binary geothermal plant**
- Is the most important safety device o the power boiler.
 - **Safety Valve**
- During hydrostatic test, the safety valves should be
 - **Removed**

- Where deaerating heaters are not employed, it is recommended that the temperature of the feed water be not less than _____.
 - **197 deg C**
- Is a reaction during which chemical energy is released in the form of heat.
 - **Endothermic reaction**
- By reheating the steam in an ideal Rankine cycle the moisture content at boiler exit will
 - **Increase**
- Presently the highest steam temperature allowed at the turbine inlet is about _____.
 - **620 deg C**
- Two most common gases employed in Stirling and Ericsson cycles are
 - **Hydrogen and Helium**
- In most common designs of Gas turbine, the pressure ratio ranges from
 - **11 to 16**
- The fuel injection process in diesel engine starts when the piston _____.
 - **Approaches TDC**
- In the cut - off ratio of diesel engine cycle increases the cycle efficiency will
 - **Decrease**
- The fuel used in a power plant that is used during peak periods
 - **Liquid**
- Typical compression ratio of Otto cycle is
 - **8**
- Joule Thompson coefficient is equal to zero, then the process will become
 - **Isothermal**
- In the fluid passed through a nozzle its entropy will:
 - **Remains the same**
- Refrigerants consisting of mixtures of two or more different chemical compounds, often used individually as refrigerants for other applications
 - **Blends**
- Pairs of mating stop valves that allows sections of a system to be joined before

opening these valves or separated after closing them

- **Companion Valve**
- An enclosed passage way that limits travels to a single path
 - **Corridor**
- For immediate dangerous to life or health (IDLH), the maximum concentrating from which unprotected persons are able to escape within ____ without escape-impairing symptoms or irreversible health.
 - **30 minutes**
- The volume as determined from internal dimensions of the container with no allowance for the volume of internal parts.
 - **Internal Gross Volume**
- A waiting room or large hallway serving as a waiting room.
 - **Lobby**
- A continuous and unobstructed path of travel of any point in a building or structure to a public way.
 - **Mean Of Egress**
- Any device or portion of the equipment used to increase refrigerant pressure.
 - **Pressure-Imposing Element**
- The quantity of refrigerant stored at some point is the refrigeration system for operational, service, or standby purposes.
 - **Pump Down Charge**
- Secondary refrigerant is a liquid use for the transmission of heat, without change of state, and having no flash point or a flash point above ____ as determined from ASTM.
 - **150 degrees F**
- A service valve for dual pressure-relief devices that allows using one device while isolating the other from the system, maintaining one valve in operation at all times.
 - **Three-Way Valve**
- Tubing that is unenclosed and therefore exposed to crushing, abrasion, puncture, or similar damage after installation.
 - **Unprotected Tubing**
- Refers to blend comprising multiple components of different volatile that, when used in refrigeration cycles, change volumetric composition and saturation temperature as they

evaporate (boil) or condense at constant pressure.

- **Zeotropic**
- Is a premises or that portion of premise from which, because they are disabled, debilitated, or confined, occupants cannot readily leave without the assistance of others.
- **Institutional Occupancy**
- Is one of which a secondary coolant is in direct contact with the air or other substance to be cooled or heated.
- **Indirect Closed System**
- Refrigerant number R-744 is:
 - **Carbon Dioxide**
- Refrigerant number R-1150 is:
 - **Ethene**
- Refrigerant number R-40 is:
 - **Chloromethane**
- When the air duct system serves several enclosed spaces, the permissible quantity of refrigerant in the system shall not exceed the amount determined by using the total volume of those spaces in which the airflow cannot be reduced to less than ____ of its maximum when the fan is operating.
- **One-Quarter**
- The space above a suspended ceiling shall not be included in calculating the permissible quantity of refrigerant in the system unless such space is continuous and is part of the air return system.
- **Plenums**
- Which with of the following is not possible location of service valve?
 - **Outlet Of Condenser**
- A coil in series with evaporator that is use to prevent the liquid refrigerant entering the compressor.
- **Drier Loop**
- A type of valve connected from discharge of compressor directly to suction that is normally closed and will open automatically only if there is high discharge pressure.
- **Solenoid Valve**
- Use to increase the capacity of condenser.
 - **Desuperheating Coils**
- Is use to subcooled the refrigerant from the condenser.
 - **Liquid Subcooler**
- Which of the following is NOT a part of low pressure side of the refrigeration system?
 - **Liquid Line**
- Which of the following is NOT a part of high pressure side of the refrigeration system?
 - **Suction Line**
- sub cooling the refrigerant in refrigeration system, the compressor power per unit mass will
 - **Remains the same**
- Superheating the refrigerant in refrigeration system, the specific volume t compressor suction will
 - **Remains the same**
- By subcooling the refrigerant in refrigeration system, the specific volume at compressor suction will
 - **Remains the same**
- Which of the following is NOT a type of air-cooled condenser?
 - **Shell and Tube**
- A type of refrigerant control typically used in household refrigeration.
 - **Capillary Tube**
- Type of condenser that operates like a cooling tower.
 - **Evaporative Condenser**
- The major problem of heat pump is
 - **Frosting**
- Dominant refrigerant used in commercial refrigeration system.
 - **R-502**
- Cascade refrigeration system are connected in
 - **Series**
- Is used to heat up the solution partially before entering the generator in absorption refrigeration system.
 - **Regenerator**

- The COP of actual absorption refrigeration system is usually.
 - **Less Than 1**
- Sight glass is often located at:
 - **Liquid Line**
- Use to detects vibration in current caused by the ionization of decomposed refrigerant between two opposite-charged platinum electrodes.
 - **Electronic Detector**
- The ability of oil to mix with refrigerants
 - **Miscibility**
 -
- Joint and all refrigerants containing parts of a refrigerant system located in an air duct carrying conditioned air to and from an occupied space shall be constructed to withstand a temperature of _____ without leakage into the airstream.
 - **700°F**
 -
- Refrigerant piping crossing an open space that affords passageway I any building shall be not less than _____ above the floor unless the piping is located against the ceiling of such space and is permitted by the authority having jurisdiction.
 - **2.2m**
- Methyl chloride shall not be in contact with
 - a. aluminum
 - b. zinc
 - c. magnesium
 - d. **All of the above**
- Shall not be in contact with any halogenated refrigerants
 - **Magnesium**
- Are suitable for use if ammonia system
 - **Aluminum and its alloy**
- If a pressure-relief device is used to protect a pressure vessel having an inside dimension of 6in or less, the ultimate strength of the pressure vessel so protected shall be sufficient to withstand a pressure at least _____ the design pressure
 - **3 times**
- Seats and disc shall be limited in distortion, by pressure or other cause, to a set pressure change of not more than _____ in a span of five years.
 - **5%**
- Liquid receivers, if used, or parts of a system designed to receive the refrigerant charge during pump down charge. The liquid shall not more than _____ of the volume when temperature of the refrigerant is 90°F
 - **90%**
- The discharge line(B4) shall be vented to the atmosphere through a _____ fitted to its upper extremity.
 - **Diffuser**
- Convert fossil fuel into shaft work.
 - **Dendrothermal power plant**
- Ultimate strength drops by 30% as steam temperature raises from _____ for unalloyed steel.
 - **400 to 500°C**
- Recent practice limits steam upon condensation gets subcooled.
 - **Drain cooler**
- Needs only single pump regardless of number of heaters
 - **Closed heater**
- Is also known as daerator
 - **Open heater**
- Dissolve gases like _____ makes water corrosive react with metal to form iron oxide
 - **O₂ and N₂**
- A cycle typically used in paper mills, textile mill, chemical factories, sugar factories and rice mills.
 - **Cogeneration cycle**
- Type of turbine employed where steam continuously extracted for process heating
 - **Passout turbine**
- Which of the following is used for Binary cycle power generation for high temperature application
 - a. mercury
 - b. sodium
 - c. potassium
 - d. **All the above**
- Critical temperature of mercury is
 - **1460°C**
- Critical pressure of mercury is

- **108 Mpa**
- Method used in converting heat directly to electricity by magnetism
 - **Magnetodynamic**
- Which of the following NOT a material used for thermoelectric elements
 - **Zinc telluride**
- A type of coal formed after anthracite
 - **Graphite**
- Which of the following has the lowest grade of coal?
 - **Lignite**
- Which of the following helps in the ignition of coal?
 - **Volatile matter**
- Is the ratio of fixed carbon and volatile matter
 - **Fuel ratio**
- A suspension of a finely divide fluid in another
 - **Emulsion**
- Contains 90% gasoline and 10% ethanol
 - **Gasohol**
- Process used commercially in coal liquefaction
 - **Fisher-tropsch process**
- Is an organic matter produced by plants in both land and water
 - **Biomass**
- In thermal power plant, forced draft fans are located at the
 - **Foot of the stack**
- In thermal power plant, forced draft fans are installed at the
 - **Inlet of the preheated**
- Known as the drumless boiler
 - **Once-through boiler**
- Reduces the steam temperature by spraying low temperature water from boiler drum
 - **Desuperheater**
- Carbon dioxide can be removed by
 - **Aerator**
- Is often used to absorb silica from water
 - **Magnesium hydroxide**
- Presence of excess hydrogen ions makes the water
 - **Acidic**
- Ph value of _____ is usually maintained for boiler water to minimized corrosion
 - **10.5**
- What type of turbine that has a degree of reaction of $\frac{1}{2}$?
 - **Rarsons turbine**
- The cooling water is made to fall in series of baffles to expose large surface area steam fed from below to become in direct contact.
 - **Barometric condenser**
- Show the variation of river flow(discharge) with time
 - **Hydrograph**
- Is an open channel erected on a surface above the ground
 - **Flume**
- Type of turbine used up to 300m head
 - **Deriaz turbine**
- A turbine that has a diagonal flow
 - **Deriaz turbine**
- Oil is atomized either by air blast or pressure jet at about
 - **70 bar**
- Type of solid injection that use single pump supplies fuels under high pressure to a fuel header
 - **Common rail injection**
- Water flow in diesel engine that is caused by density differential.
 - **Thermosiphon cooling**
- Type of lubrication system in diesel engine in which oil from pump is carried to a separate storage tank outside the engine cylinder and used for high capacity engine
 - **Dry sump lubrication system**

- Produces extreme pressure differentials and violent gas vibration
 - **Detonation**
- In a spark ignition engine, the detonation occurs near the_____
 - **End of combustion**
- In a compression ignition engine, the detonation occurs near the_____
 - **Beginning of combustion**
- Morse test is use to measure the_____ of multi-cylinder engine
 - **Indicated power**
- Ignition delay can be minimized by adding_____ to decrease engine knocking.
 - **Ethyl nitrate**
- For the submerged plane surface, the point on the surface where the resultant force acts is called the
 - **Center of pressure**
- At any point in fluid at rest, the pressure is the same in all directions. This principle is known as:
 - **Pascal's Law**
- The hot wire manometer is used to measure
 - **Gas velocities**
- The pitot static tube measures
 - **The static pressure**
- The terminal velocity of a small sphere setting in a viscous fluid varies as the
 - **Inverse fluid viscosity**
- Pressure drag results from
 - **Skin friction**
- The pressure coefficient is the ratio of pressure forces to
 - **Inertia forces**
- Is the condition of pressure and temperature at which a liquid and its vapor are indistinguishable
 - a. Relative humidity
 - b. Absolute humidity
 - c. Critical point
 - d. Dew point
- Ignition of the air fuel mixture in the intake of the exhaust manifold:
 - a. Backlash
 - b. Backfire
 - c. Exhaust pressure
 - d. Back pressure
- The energy of a fluid flowing at any section in a pipeline is a function of:
 - a. velocity of flow only
 - b. pressure only
 - c. Height above a chosen datum, density, internal energy, pressure and velocity of flow
 - d. Pressure, height above a chosen datum, velocity of flow, density of fluid
- The number of protons in the nucleus of an atom of the number of electrons in the orbit of an atom:
 - a. Atomic volume
 - b. Atomic number
 - c. Atomic weight
 - d. Atomic mass
- In a P-T diagram of a pure substance, the curve separating the solid phase from the liquid phase is:
 - a. vaporization curve
 - b. Fusion curve
 - c. Boiling point
 - d. Sublimation point
- Form of energy associated with the kinetic energy of the random motion of large number of molecules:
 - a. Internal energy
 - b. Kinetic energy
 - c. Heat of fusion
 - d. Heat
- A Francis Turbine has what flow?
 - a. Inward flow reaction
 - b. Outward flow reaction
 - c. Outward flow impulse
 - d. Inward flow impulse
- Is the condition of pressure and temperature at which liquid and its vapor are indistinguishable:
 - a. critical point
 - b. Dew point
 - c. Absolute humidity
 - d. Relative humidity
- When a substance in gaseous state is below its critical temperature, it is called:
 - a. vapor
 - b. cloud
 - c. moisture
 - d. steam
- The final horsepower delivered to the equipment:
 - a. brake horsepower
 - b. effective horsepower
 - c. indicated horsepower
 - d. friction horsepower

- Pump used to increased air pressure above normal, air is then used as a motive power:
 - a. Air cooled engine
 - b. Sub-cooled liquid
 - c. Air condenser
 - d. Air injection
- In sensible cooling process, the moisture content:
 - a. does not change
 - b. decrease
 - c. indeterminate
 - d. increases
- The dividing point between the high-pressure and low pressure sides of the refrigerating cycle occurs at the:
 - a. Expansion valve
 - b. Compressor
 - c. Condenser
 - d. Cooking coil
- What is the force which tends to draw a body toward the center about which it is rotating?
 - a. Centrifugal force
 - b. Centrifugal in motion
 - c. Centrifugal advance
 - d. Centripetal force
- A simultaneous generation of electricity and steam (or heat) in a single power plant:
 - a. Steam turbine-gas turbine plant
 - b. Cogeneration
 - c. Gas turbine plant
 - d. Waste heat recovery
- Is one whose pressure is higher than the saturation pressure corresponding to its temperature:
 - a. Saturated pressure
 - b. Compressed liquid
 - c. Saturated liquid
 - d. Compressed gases
- If the fluid travels parallel to the adjacent layers and the paths of individual particles do not cross, the flow is said to be:
 - a. turbulent
 - b. critical
 - c. dynamic
 - d. laminar
- What is the apparatus used in the analysis of combustible gases?
 - a. Calorimeter differential
 - b. Calorimeter gas
 - c. Calorimetry
 - d. Calorimeter
- What is the clockwork-operated device which records continuously the humidity of the atmosphere?
 - a. Hectograph
 - b. Hydrometer
 - c. Hygrograph
 - d. Hydrograph
- The ratio of the average load to the peak load over a designated period of time is called:
 - a. Load factor
 - b. Reactive factor
 - c. Diversity factor
 - d. Plant use factor
- When a system deviates infinitesimally from equilibrium at every instant of its state, it is undergoing
 - a. Isobaric process
 - b. Quasi-static process
 - c. Isometric process
 - d. Cyclic process
- Heat transfer due to density differential:
 - a. convection
 - b. nuclear
 - c. Conduction
 - d. radiation
- Work done per unit charge when charge is moved from one point to another:
 - a. Equipotential surface
 - b. Potential at a point
 - c. Electrostatic unit
 - d. Potential difference
- What amount of air is required in a low by pass factor?
 - a. greater
 - b. lesser
 - c. indeterminate
 - d. does not change
- Percent excess air is the difference between the air actually supplied and the theoretically required divided by:
 - a. the theoretically air required
 - b. the deficiency air supplied
 - c. the actually air supplied
 - d. None of these
- The sum of the energies of all molecules in a system, energies appear in several complex forms, is the
 - a. Kinetic energy
 - b. Potential energy
 - c. Internal energy
 - d. Frictional energy
- At any instant, the number of particles passing every cross-section of the stream is the same, the flow is said to be
 - a. steady flow
 - b. Uniform flow
 - c. Continuous flow
 - d. Turbulent flow
- The ratio of the density of a substance to the density of some standard substance is called
 - a. relative density
 - b. specific gravity
 - c. specific density
 - d. Relative gravity
- The total energy in a compressible or incompressible fluid flowing across and suction in a pipeline is a function of:
 - a. pressure and velocity
 - b. pressure, density and velocity

- c. pressure, density and velocity, and viscosity
 - d. flow energy, kinetic energy, height above datum and internal energy
- The locus or elevations to which water will rise in the piezometer tube is termed
 - a. Energy gradient
 - b. Friction head
 - c. Hydraulic gradient
 - d. Critical depth
- The heat per unit mass per degree change in temperature is called:
 - a. Heat fusion
 - b. Heat of vaporization
 - c. Relative heat
 - d. Specific heat
- The temperature to which the air must be cooled at constant pressure to produce saturation is called:
 - a. Boiling point
 - b. Dew point
 - c. Critical point
 - d. Triple point
- The mass of water vapor per unit volume of air
 - a. absolute humidity
 - b. specific humidity
 - c. relative humidity
 - d. critical humidity
- The heat transfer due to motion of matter caused by a change in density is called:
 - a. radiation
 - b. convection
 - c. conduction
 - d. absorption
- When the expansion or compression of gas takes place without transfer of heat to or from the gas, the process is called:
 - a. isothermal
 - b. isentropic
 - c. adiabatic
 - d. reversible
- The external pressure applied to a confined fluid increases the pressure of every point in the fluid by an amount equal to the external pressure. This is known as:
 - a. Archimedes Principle
 - b. Torricelli's Theorem
 - c. Bernoulli's Theorem
 - d. Pascal's Law
- Weight per unit volume is termed as:
 - a. Specific gravity
 - b. Density
 - c. Weight density
 - d. Pressure
- The speed with which a liquid escapes from a vessel through an orifice is given by:
 - a. Archimedes Principle
 - b. Torricelli's Theorem
 - c. Bernoulli's Theorem
 - d. Flow equation
- When gas is heated at constant volume, the process is called:
 - a. isotropic
 - b. isobaric
 - c. isometric
 - d. adiabatic
- When gas is heated at constant pressure, the process is called:
 - a. isometric
 - b. adiabatic
 - c. polytropic
 - d. isobaric
- An instrument that measures density
 - a. Manometer
 - b. Hydrometer
 - c. Barometer
 - d. Hydraulic meter
- This law states that pressure applied at a point in a confined liquid is transmitted equally to all other points.
 - a. Boyle's Law
 - b. Charles' Law
 - c. Len's Law
 - d. Pascal's Law
- The temperature at which its vapor pressure is equal to the pressure exerted on the liquid
 - a. Absolute humidity
 - b. Calorimeter
 - c. Boiling point
 - d. Thermal energy
- Type of turbine that has high pressure and low pressure is called
 - a. compound turbine
 - b. gas turbine
 - c. impulse turbine
 - d. compound turbine
- The design of an air supply duct of an air conditioning system
 - a. adds moisture to the air
 - b. Lowers the temperature of the air
 - c. Does not affect the distribution of air
 - d. Affects the distribution of air
- The changing of solid directly to vapor, without passing through the liquid state is called:
 - a. Evaporation
 - b. vaporization
 - c. Sublimation
 - d. Condensation
- The volume of a liquid passing a cross section of a stream in unit time is called:
 - a. steady flow
 - b. uniform flow
 - c. discharge
 - d. continuous flow
- If the volume of the confined gas is constant, the pressure is directly proportional to the absolute temperature. This is known as:
 - a. Kelvin's Law
 - b. Boyle's Law

- c. Charles' Law
 - d. Joule's Law
- If the temperature of a confined gas is constant, the product of the pressure and volume is constant. This is known as:
 - a. Kelvin's Law
 - b. Boyle's Law
 - c. Charles' Law
 - d. Joule's Law
- At the same pressure and temperature equal volumes of all gases contain equal number of molecules, this is known as:
 - a. Faraday's Law
 - b. Boyle's Law
 - c. Len's Law
 - d. Avogadro's Law
- Amount of heat required to change the temperature of 1 g of water 1°C
 - a. British thermal unit
 - b. Joules
 - c. Calorie
 - d. Heat
- Evaluation of equipment performance requires
 - a. constant supervision
 - b. daily log sheet
 - c. manufacturer's representative
 - d. all of these
- Atmospheric pressure at sea level is:
 - a. 76 mm Hg
 - b. 29 ft H₂O
 - c. 14.7 psi
 - d. 101.325 kg/cm²
- Soot on tubes will result in:
 - a. loss of efficiency
 - b. loss of steam
 - c. loss of water
 - d. lose of fire
- Fuel oil mechanical atomization depends on:
 - a. viscosity
 - b. pressure
 - c. temperature
 - d. volume
- Air contains by weight:
 - a. 23% O₂
 - b. 21% O₂
 - c. 22% O₂
 - d. 20% O₂
- An instrument used to determine the specific gravity of a solution
 - a. Tachometer
 - b. Sling psychrometer
 - c. Hygrometer
 - d. Hydrometer
- A boiler in which the products of combustion pass through the tube which is surrounded by water.
 - a. water-tube boiler
 - b. vertical-tube boiler
 - c. fire-tube boiler
 - d. horizontal-tube boiler
- Work done in pushing a fluid across a boundary, usually into or out of a system
 - a. torque
 - b. partial pressure
 - c. kinetic energy
 - d. flow energy
- If heat energy a gas at the bottom, the hotter gas expands, gets lighter per unit volume and begins to be replaced by gravitational movements called:
 - a. free electrons
 - b. isobaric process
 - c. free convection
 - d. isothermal process
- This heated air, being lighter, rises and circulates through the house or is forced through by a fan is called
 - a. vacuum
 - b. vapor
 - c. steam
 - d. forced convection
- Is a chart on which enthalpy is the ordinate and entropy the abscissa.
 - a. Ericson cycle
 - b. Mollier Chart
 - c. Charles' Law
 - d. Bernoulli's Chart
- Is one with rigid boundaries exchanging neither energy nor mass with its surroundings
 - a. Entropy
 - b. ideal gases
 - c. isolated system
 - d. heat
- An unbalanced one in search of an equilibrium state or a steady state
 - a. ram effects
 - b. radiosity
 - c. equilibrium flow
 - d. transient process
- The term velocity head is called:
 - a. friction head
 - b. head loss
 - c. fluid friction
 - d. pressure head
- When system is in the thermodynamic equilibrium with the natural environment it is in:
 - a. dead state
 - b. sink
 - c. equilibrium
 - d. neutral
- A body wholly or partly immersed on a fluid is buoyed up by a force equal to the weight of the fluid it displaces. This is known as:
 - a. Bernoulli's Theorem
 - b. Torrecelli's Theorem
 - c. Pascal's Theorem
 - d. None of the above

- The ratio of the sum of individual maximum demands of the system to the total demand of the whole system:
 - a. demand factor
 - b. Power factor
 - c. Utilization
 - d. diversity factor
- The ratio of the average load over the designated period of time to peak load in that period of time
 - a. Diversity factor
 - b. Plant use factor
 - c. Capacity factor
 - d. None of the above
- The ratio of kWh generated to the product of the capacity of the plant in kW to the total number of hours the plant has been in actual use.
 - a. Diversity factor
 - b. Utilization factor
 - c. Load factor
 - d. Demand factor
- A type of water turbine where a jet of water is made to fall on the blades or buckets and due to the impulse of the water, the turbine starts moving
 - a. Pelton wheel
 - b. Reaction turbine
 - c. Francis turbine
 - d. Kaplan turbine
- The sum of the pressure head, elevation head, and the velocity head remains constant. This is known as:
 - a. Torrecelli's Theorem
 - b. Archimedes Principle
 - c. Bernoulli's Theorem
 - d. Boyle's Law
- Is the attraction between like molecules
 - a. Absorption
 - b. adhesion
 - c. diffusion
 - d. cohesion
- Is the attraction between unlike molecules
 - a. diffusion
 - b. cohesion
 - c. extraction
 - d. adhesion
- The process of one substance mixing with another because of molecular motion
 - a. saturation
 - b. absorption
 - c. adhesion
 - d. diffusion
- States that a heat engine cannot transfer heat from a body to another at a higher temperature unless external energy is supplied to the engine
 - a. First Law of thermodynamics
 - b. Planck's Law
 - c. Second Law of thermodynamics
 - d. Third Law of thermodynamics
- The escape of burned gases from the combustion chamber through the piston rings and into the crankcase:
 - a. blowby
 - b. priming
 - c. scavenging
 - d. foaming
- The type of heat that changes the physical state of a substance
 - a. specific heat
 - b. latent heat
 - c. sensible heat
 - d. over heat
- Water containing a high proportion of chemicals is:
 - a. Potable
 - b. Treated
 - c. Soft
 - d. Hard
- A thermocouple is another name of:
 - a. two conductors
 - b. two insulators
 - c. two thermometers
 - d. two thermostats
- The first act when taking over boiler watch is to:
 - a. check the system
 - b. clean strainers
 - c. check the water in the boiler
 - d. check the boiler feed pump
- Heat transfer by contact is called:
 - a. Convection
 - b. conduction
 - c. radiation
 - d. none of the above
- The part of the boiler designed for burning the fuel.
 - a. chimney
 - b. tube
 - c. gauge cock
 - d. furnace
- The amount of heat passing through a body of 1 m² cross section and 1 m thick in 1 hour at a temperature difference of 1°C.
 - a. quantity of heat
 - b. Specific heat
 - c. Latent heat of solid
 - d. Thermal conductivity
- Is a substance that exists, or is regarded as existing, as a continuum characterized by low resistance to flow and the tendency to assume the shape of its container.
 - a. fluid
 - b. gas
 - c. ice
 - d. volume
- The weight of a body means the
 - a. mass of a body
 - b. volume of a body
 - c. force of gravity of a body

- d. molecular weight
- The negative sign is for a gage reading called:
 - a. transmission of pressure
 - b. vapor pressure
 - c. pressure head
 - d. vacuum pressure
- When property changes cease, the bodies are said to be in:
 - a. thermal equilibrium
 - b. slug
 - c. change in volume
 - d. change in pressure
- When a certain mass of fluid in a particular state passes through a series of processes and returns to its initial state, it undergoes a
 - a. Cycle
 - b. reversible non-flow process
 - c. irreversible process
 - d. equilibrium
- Is an open system in which there is no change of stored mass
 - a. cooling system
 - b. steady flow system
 - c. dew point
 - d. elastic system
- Work per unit time is:
 - a. specific heat
 - b. torque
 - c. specific mass
 - d. power
- The number of molecules in a mole of any substance is a constant called
 - a. Thompson's constant
 - b. Rankine cycle
 - c. Otto cycle
 - d. Avogadro's number
- The gas constant per molecule, called
 - a. Archimedes constant
 - b. Mollier's constant
 - c. Avogadro's constant
 - d. Boltzmann's constant
- The kind of process, as the opening of water faucet and letting the water flow, is called
 - a. Throttling process
 - b. torrecelli's constant
 - c. Orifice
 - d. Pressure and velocity
- "At absolute zero, the entropy of a pure substance (in equilibrium at 0°R) in some "perfect" crystalline form becomes zero"
 - a. First Law of Thermodynamics
 - b. Third Law of Thermodynamics
 - c. Second Law of Thermodynamics
 - d. Fourth Law of Thermodynamics
- Form of energy, associated with the kinetic energy of the random motion of large numbers of molecules
 - a. Heat of fusion
 - b. Kinetic energy
 - c. Internal Energy
 - d. Heat
- Which property of fluids is of fundamental importance in the study of hydraulics?
 - a. Unit weight
 - b. Mass density
 - c. Specific gravity
 - d. All of the above
- Summation of all heads in one section is equal to the summation of all heads in another section.
 - a. Archimede's Principle
 - b. Bernoulli's Principle
 - c. Torrecelli's Principle
 - d. Boyle's Law
- Absolute viscosity of a fluid varies with pressure and temperature and is defined as a function of:
 - a. Density and angular deformation rate
 - b. Density, shear stress and angular deformation rate
 - c. Density and shear stress
 - d. Shear stress and angular deformation rate
- The hydraulic formula $CA\sqrt{2gh}$ is used to find the
 - a. Quantity of the discharge through an orifice
 - b. Velocity of flow in a closed conduit
 - c. Length of pipe in a closed network
 - d. Friction factor of pipe
- Entropy is the measure of
 - a. The change of enthalpy of steam C
 - b. Randomness of disorder
 - c. The internal energy of gas
 - d. The heat capacity of a substance
- Materials which are capable of absorbing moisture:
 - a. Refractories
 - b. hygroscopic
 - c. polymer
 - d. all of the above
- A vessel permanently connected to a system by inlet and outlet for storage of liquid refrigerant
 - a. refrigerant container
 - b. flash tank
 - c. liquid receiver
 - d. surge tank
- Are industrial equipments used to remove relatively small amount of water or moisture from large amount of solids into a gas or air stream
 - a. Continuous driers
 - b. current driers
 - c. batch driers
 - d. all of the above
- Ice plants are rated in terms of:

- a. tons of refrigeration
 - b. coefficient of performance
 - c. tons of ice per day
 - d. all of the above
- Flexible material used to seal pressurized fluids, normally under dynamic conditions.
 - a. packing
 - b. polymer
 - c. gasket
 - d. seal
- An instrument that determines the heating value of fuel.
 - a. Calorimeter
 - b. hydrometer
 - c. psychrometer
 - d. tachometer
- A nuclear power reactor where heavy fuel atoms (U233 or U235) are split.
 - a. fusion reactor
 - b. fission reactor
 - c. solid reactor
 - d. none of these
- A chemical substance used for softening feedwater.
 - a. activated carbon
 - b. sulfur
 - c. zeolite
 - d. magnesium
- In fluid flow, if the fluid travels parallel to the adjacent layers and the paths of the individual particles do not cross, the flow is said to be:
 - a. uniform
 - b. turbulent
 - c. dynamic
 - d. laminar
- An air or gas handling unit which removes air:
 - a. Blower
 - b. Fan
 - c. Compressor
 - d. All of the above
- Type of coal which is more widely used for firing boilers
 - a. lignite
 - b. bituminous
 - c. anthracite
 - d. peat
- Term referring to the products of combustion
 - a. exhaust gas
 - b. flue gas
 - c. smoke
 - d. stack gas
- Refers to the temperature at which all molecular motion ceases according to the kinetic theory of heat.
 - a. flash point
 - b. Critical point
 - c. Saturation point
 - d. absolute zero
- Heating fuel oil will reduce its:
 - a. volume
 - b. viscosity
 - c. valence
 - d. pressure
- A pump that requires an internal or external relief is a:
 - a. centrifugal pump
 - b. gear
 - c. steam pump
 - d. reciprocation pump
- To remove suspended solids from water, you need:
 - a. filtration
 - b. atomization
 - c. precipitation
 - d. demineralization
- The soap solution test is recorded in:
 - a. cm
 - b. ppm
 - c. mg/l
 - d. mg/cm³
- If you cannot find water in a boiler, you should:
 - a. add more water
 - b. call a supervisor
 - c. Shut down boiler
 - d. run away immediately
- The change of the state of a substance from gaseous to the liquid form
 - a. condensation
 - b. sublimation
 - c. evaporation
 - d. fission
- Performance of cooling towers is commonly expressed in terms of:
 - a. air and water pressure
 - b. air and water temperature
 - c. range and approach
 - d. cooling draft
- In testing halocarbon refrigerant leaks:
 - a. A soap solution and water is used
 - b. sulfur candle test is used
 - c. halide torch is used
 - d. a digital meter is used
- A receiver in an air compressor system is used to:
 - a. avoid cooling air before using
 - b. reduce the work needed during the compression
 - c. collect the water and grease suspended in air
 - d. remove the air discharge pressure
- One ton of refrigeration is a heat equivalent to:
 - a. 2413 BTU/hr
 - b. 2545 BTU/hr
 - c. 990 BTU/hr
 - d. 12000 BTU/hr
- The science of a low temperature which deals with the liquefaction of gases

- a. air conditioning
 - b. ventilation
 - c. cryogenics
 - d. psychrometry
- Is one whose temperature is below the saturated temperature of one gram of water to one degree centigrade.
 - a. compressed
 - b. condensed
 - c. superheated
 - d. subcooled
- Based on PSME code, for piping identification, water pipes should be painted with what color?
 - a. yellow
 - b. white
 - c. green
 - d. black
- The quantity of heat required to raise the temperature of one gram of water to one degree centigrade.
 - a. Btu
 - b. Calorie
 - c. Joule
 - d. ft-lb
- A fluid in which energy can be stored and from which energy can be removed
 - a. working substance
 - b. liquids
 - c. gases
 - d. furnace
- The temperature scale wherein ice, liquid and vapor coexist is:
 - a. boiling point
 - b. triple point
 - c. flash point
 - d. critical point
- A property which is defined as the mass divided by the volume
 - a. mass density
 - b. specific gravity
 - c. specific volume
 - d. relative density
- The ability to do work is:
 - a. power
 - b. force
 - c. energy
 - d. heat
- Weight per unit volume is termed as:
 - a. specific gravity
 - b. weight density
 - c. specific density
 - d. pressure
- What do you take by multiplying the force or weight by distance?
 - a. Power
 - b. Energy
 - c. Work
 - d. Horsepower
- A positive displacement pump needs
 - a. a strong motor
 - b. a relief valve
 - c. purging
 - d. feedwater
- Anthracite is mostly:
 - a. hydrogen
 - b. diamond
 - c. graphite
 - d. carbon
- The ratio of the refrigerating effect to the work of compression
 - a. coefficient of performance
 - b. thermal efficiency
 - c. tons of refrigeration
 - d. all of the above
- In a gas-turbine plant, its cycle is based on:
 - a. Joules or Brayton
 - b. Rankine
 - c. Ericson
 - d. Carnot
- In a compression-ignition or Diesel cycle, the processes are:
 - a. SVSV
 - b. SPSV
 - c. SPSP
 - d. PSPS
- Which one has the least generation cost among the following:
 - a. Solar Power Plant
 - b. Diesel Power Plant
 - c. Geothermal Power Plant
 - d. Hydroelectric Power Plant
- Most commonly used condenser type in industries
 - a. surface type
 - b. contact type
 - c. mixed type
 - d. submerged type
- Temperature at which oil burns continuously is called:
 - a. flash point
 - b. pour point
 - c. fire point
 - d. none of these
- It connects the turbine runner with the tail race water to prevent any loss of head due to the elevation of the turbine wheel.
 - a. bucket
 - b. draft tube
 - c. water tube
 - d. tail race
- The process of the substance mixing with another because of molecular motion
 - a. cohesion
 - b. absorption
 - c. adhesion
 - d. diffusion
- The effect of magnesium chloride in feedwater is:
 - a. corrosion

- b. hard scale
 - c. alkalinity
 - d. electrolysis
- Strengthen and support walls of brick boiler settings:
 - a. buck stays
 - b. stay bolts
 - c. baffles
 - d. fire walls
- The carrying-over of water slugs with the steam:
 - a. convection
 - b. priming
 - c. foaming
 - d. run-off
- An air compartment maintained under pressure with connections to one or more distributing ducts
 - a. purge
 - b. plenum chamber
 - c. air receiver
 - d. air preheater
- The maximum unit strength developed at any time before rupture
 - a. yield's strength
 - b. endurance limit
 - c. ultimate strength
 - d. rupture strength
- The negative sign is for a gage reading called:
 - a. vapor pressure
 - b. ambient pressure
 - c. absolute pressure
 - d. vacuum pressure
- An automatic device which allows the passage of water but prevents the passage of steam:
 - a. steam trap
 - b. safety valve
 - c. steam power
 - d. vapor accumulator
- An apparatus for raising the temperature of boiler feedwater with exhaust steam
 - a. hotwater tank
 - b. feedwater heater
 - c. hotwell
 - d. steam heater
- The part of a refrigerating unit in which refrigerant undergoes transformation from gas to liquid
 - a. evaporator
 - b. condenser
 - c. expansion valve
 - d. compressor
- Cleaning of cylinder of exhaust gases by forcing into it a current of air which provides clearance for the next compression stroke of engine
 - a. honing
 - b. tune up
 - c. boring
 - d. scavenging
- The expansion of contraction of a gas without any transfer of heat into it or out of it:
 - a. isobaric process
 - b. isometric process
 - c. isothermal process
 - d. adiabatic process
- What is the force which tends to draw a body toward the center about which it is rotating?
 - a. centrifugal force
 - b. centripetal force
 - c. gravitational force
 - d. all of these
- Measure of hotness or coldness of the body
 - a. pressure
 - b. energy
 - c. temperature
 - d. entropy
- Low capacity pump for high pressure application:
 - a. reciprocating pumps
 - b. rotary pump
 - c. centrifugal pump
 - d. turbine pump
- The speed required to pump one gallon of water thru a head of one foot in one minute.
 - a. specific head
 - b. peripheral speed
 - c. linear speed
 - d. angular speed
- Dryer capacity is always based on the:
 - a. feed
 - b. product
 - c. moisture
 - d. bone-dry mass
- The ideal basis for refrigeration is:
 - a. vapor compression
 - b. Carnot cycle
 - c. Beau de Rochan
 - d. Reversed Carnot cycle
- Gasohols or alcogas is popularly known as:
 - a. green gasoline
 - b. LPG
 - c. Diesoline
 - d. LNG
- Among the effluents that may be released from the hydrothermal reservoir which one is the most toxic?
 - a. carbon dioxide
 - b. methane
 - c. hydrogen sulfide
 - d. ammonia
- Types of cooling tower where fan pressure is exerted at the outlet
 - a. natural draft
 - b. force draft
 - c. induce draft
 - d. mechanical draft

- Ammonia leaks in a refrigeration system maybe detected by
 - a. halide torch
 - b. sulfur candle test
 - c. oil of peppermint
 - d. acetylene torch
- The nominal capacity or the full load rating of steam turbine is:
 - a. $KVA \times P.F.$
 - b. $KW \times Ng$
 - c. $KVA \times Ng$
 - d. $KW \times P.F.$
- It is a horizontal, externally fired fire-tube boiler.
 - a. Stirling
 - b. Marine
 - c. HRT
 - d. dry-top
- A device which automatically controls the speed of an engine
 - a. accelerator
 - b. governor
 - c. tachometer
 - d. piezometer
- The temperature at which the water vapor in the air begins to condensed, or the temperature at which the relative humidity of air becomes 100%.
 - a. flash point
 - b. dew point
 - c. boiling point
 - d. freezing point
- The part of refrigerating unit in which the refrigerant vaporized to absorb heat
 - a. compressor
 - b. expansion valve
 - c. evaporator
 - d. condenser
- It consists of a wet and dry bulb thermometers mounted on a strip of metal
 - a. manometer
 - b. pyrometer
 - c. gyrometer
 - d. sling psychrometer
- If the pressure of the confined gas is constant, the volume is directly proportional to the absolute temperature
 - a. Boyle
 - b. Joule
 - c. Charles
 - d. Kelvin
- A heat exchanger through which air passes between the first and second stage of a two-stage air compressor is known as:
 - a. Aftercooler
 - b. Baudelot
 - c. Cooling tower
 - d. Intercooler
- The sum of the pressure head, elevation head, and the velocity head remains constant. This is known as:
 - a. Torrecelli's Theorem
 - b. Archimede's Principle
 - c. Bernoulli's Theorem
 - d. Boyle's Law
- Feedwater lines to a boiler must always be equipped with:
 - a. an OS and Y valve
 - b. gate valve
 - c. check valve
 - d. globe valve
- Distributes lubrication in a pump
 - a. O-ring
 - b. lantern ring
 - c. bearing
 - d. sleeve
- Dry ice is referred to:
 - a. solid carbon dioxide
 - b. solid carbon monoxide
 - c. solid carbon monoxide
 - d. solid carbon peroxide
- The most important piece of equipment in the refrigerating system:
 - a. expansion valve
 - b. evaporator
 - c. condenser
 - d. compressor
- An empty vessel which is located at the top of the penstock to store water when the turbine valve is suddenly closed.
 - a. superheated
 - b. daytank
 - c. hydrotank
 - d. surge tank
- A heat exchanger which increases the feedwater temperature using the flue gases.
 - a. preheater
 - b. economizer
 - c. deaerator
 - d. separator
- Which type of hydraulic turbine is used for medium-head hydroelectric power plants
 - a. Kaplan
 - b. Francis
 - c. Impulse
 - d. Reaction type
- Equal volumes of gases under identical conditions of pressure and temperature contain the same number of molecules.
 - a. Dalton's Law
 - b. Keppler's Law
 - c. Newton's Law
 - d. Avogadro's Law
- The most important valve on the boiler is:
 - a. safety valve
 - b. blowdown valve
 - c. feedwater valve
 - d. gate valve
- Is a curve telling a history of the flow of water from month throughout the year
 - a. hydrograph

- b. William's line
 - c. bell curve
 - d. spot curve
- A short crank that moved the valve back and forth on its seat
 - a. camshaft
 - b. eccentric
 - c. piston rod
 - d. crankshaft
- It converts mechanical energy into electrical energy
 - a. generator
 - b. motor
 - c. turbine
 - d. engine
- The kind of fuel injected to the combustion chamber of diesel engine -
 - a. atomized
 - b. vaporized
 - c. evaporized
 - d. energized
- The degrees of power interval in a 4-stroke cycle of a four-cylinder engine -
 - a. 60 degrees
 - b. 90 degrees
 - c. 180 degrees
 - d. 120 degrees
- The degrees of power interval in a 4-stroke cycle of a six-cylinder engine -
 - a. 60 degrees
 - b. 90 degrees
 - c. 180 degrees
 - d. 120 degrees
- The power interval in single-cylinder engine -
 - a. 360 degrees
 - b. 480 degrees
 - c. 560 degrees
 - d. 720 degrees
- The power interval in a 4-stroke cycle of an 8-cylinder engine
 - a. 60 degrees
 - b. 90 degrees
 - c. 180 degrees
 - d. 120 degrees
- In a 4-stroke cycle, the number of crankshaft revolutions to complete a cycle is -
 - a. one revolution
 - b. two revolutions
 - c. three revolutions
 - d. none of the above
- The stroke whereby burned gases are discharged through the upward movement of the piston of a valve which is open -
 - a. intake stroke
 - b. compression stroke
 - c. power stroke
 - d. exhaust stroke
- Which of the following is not a cause of overheating?
 - a. wrong timing
 - b. low oil level
 - c. low water level
 - d. battery overcharge
- The main purpose of a cooling system -
 - a. to cool the engine
 - b. to avoid overheating
 - c. to maintain engine-working temperature
 - d. none of the above
- The abnormal wearing-out of parts due to excessive friction -
 - a. tear
 - b. wear
 - c. score
 - d. seizure
- The main purpose of lubricating system is -
 - a. to reduce friction
 - b. to reduce noise
 - c. to reduce excessive heat
 - d. to reduce wear
- The part that provides sufficient supply of diesel fuel to the engine during operation
 - a. gasoline tank
 - b. diesel fuel reservoir
 - c. diesel fuel system
 - d. diesel fuel pump
- When both valves are close and the piston is moving upward -
 - a. intake volume
 - b. compression stroke
 - c. power stroke
 - d. exhaust stroke
- In a two-stroke cycle engine, the number of crankshaft revolutions to complete one cycle is -
 - a. one revolution
 - b. two revolution
 - c. three revolution
 - d. none of the above
- An internal combustion engine converts chemical energy to -
 - a. heat energy
 - b. mechanical energy
 - c. chemical energy
 - d. sound energy
- The assemblage of equipment that produces and delivers a flow of mechanical to electrical energy -
 - a. generator
 - b. regulator
 - c. diesel electric power plant
 - d. stationary engines
- The source of power which act as the prime mover for the generator, with horsepower ranging from 100 to 500 -
 - a. motor
 - b. gasoline engine
 - c. diesel engine

- d. steam engine
- Which valve is close during power stroke?
 - a. intake valve
 - b. exhaust valve
 - c. intake and exhaust valve
 - d. none
- Which of the following is not one of the purposes of the lubricating system?
 - a. to act as coolant
 - b. to act as sealant
 - c. to act as detergent
 - d. to act as contaminant
- Which of the following statements is correct?
 - a. Intake is more important than compression
 - b. Power is the most important stroke
 - c. Exhaust is next to power in importance
 - d. Compression occurs when the fuel charge is compressed into smaller volume
- Which of the following is not a property of lubricating oil?
 - a. carbon residue
 - b. flash point
 - c. acidity
 - d. combustion
- Pressure filter that builds up working pressure is -
 - a. 100 psi
 - b. 125 psi
 - c. 150 psi
 - d. 200 psi
- The type of cooler extensively used for medium and large size engines -
 - a. plate cooler
 - b. radiator cooler
 - c. shell and tube cooler
 - d. disk cooler
- The type of pump where oil is being carried away by two vanes during operation
 - a. gear type
 - b. rotary type
 - c. lobe type
 - d. oil pump
- Which of the following is not a part of the air intake system?
 - a. blower with supercharging
 - b. thermometer
 - c. air intake silencer
 - d. air filter with piping connection
- Which of the following engines use vane blowers?
 - a. aircraft engines
 - b. marine engines
 - c. locomotive
 - d. all of the above
- Hand-starting is used only for small engines having a bore diameter not exceeding
 - a. 2 inches
 - b. 4 inches
 - c. 3 inches
- d. 6 inches
- Which of the following requires large filter to handle oil for a medium-sized engine?
 - a. continuous filtering
 - b. shunt filtering
 - c. bypass filtering
 - d. hatch filtering
- The function of an unloader on an electric motor-driven compressor is to:
 - a. reduce the speed of the motor when the maximum pressure is reached.
 - b. Drain the condensate from the cylinder
 - c. Release the pressure in the cylinders in order to reduce the starting load.
 - d. Prevent excess pressure in the receiver
- The power required to deliver a given quantity of fluid against a given head with no losses in the pump is called:
 - a. wheel power
 - b. brake power
 - c. hydraulic power
 - d. indicated power
- An engine indicator is generally used to measure
 - a. steam temperature
 - b. heat losses
 - c. steam cylinder pressure
 - d. errors to gauge reading
- Measure of ability of a boiler to transfer the heat given by the furnace to the water and steam is:
 - a. grate efficiency
 - b. stoker efficiency
 - c. furnace efficiency
 - d. boiler efficiency
- How do you describe a non-flow process where in the volume remains constant?
 - a. isometric
 - b. isentropic
 - c. isobaric
 - d. isenthalpic
- In the processing section, there is an instrument frequently used to measure the flow rate of fluids. What is the instrument consisting of a venical passage with variable cross-sectional area, a float and a calibrated scale?
 - a. rotameter
 - b. pitot-tube
 - c. rota aire
 - d. manometer
- The law states that entropy of all perfect crystalline solids is zero at absolute zero temperature
 - a. Newton's Law
 - b. Third Law of Thermodynamics
 - c. First Law of Thermodynamics
 - d. Second Law of Thermodynamics
- A goose-neck is installed in the line connecting a steam gauge to a boiler to:

- a. maintain constant steam flow
 - b. protect the gauge element
 - c. prevent steam knocking
 - d. maintain steam pressure
- A heat transfer device that reduces a thermodynamic fluid from its vapor phase to its liquid phase such as in a vapor-compression refrigeration plant or in a condensing steam power plant.
 - a. flash vessel
 - b. cooling tower
 - c. condenser
 - d. steam separator
- President F. V. Ramos approved on February 12, 1998 a Republic Act which is an act to regulate the practice of Mechanical Engineering in the Philippines, otherwise known as the new M.E. Law. What is this act?
 - a. RA No. 9645
 - b. RA No. 8495
 - c. RA No. 8594
 - d. RA No. 8945
- A manometer is an instrument that is used to measure:
 - a. air pressure
 - b. heat radiation
 - c. Condensate water level
 - d. air volume
- What is the color code for air pressure?
 - a. red
 - b. light blue
 - c. brown
 - d. violet
- The carbon dioxide percentage in the flue gas of an efficiently fired boiler should be approximately
 - a. 15%
 - b. 12%
 - c. 16%
 - d. 20%
- Which material should be used for the construction of a ducting system?
 - a. steel
 - b. iron
 - c. Aluminum
 - d. All of these
- An ideal fluid is one that
 - a. is very viscous
 - b. obeys Newton's Law of Viscosity
 - c. None of the above
 - d. frictionless and is compressible
- In industrial plant design, in calculating the working space requirement, the maximum number of persons working or will be working shall not exceed one person per _____ cubic meter.
 - a. 12
 - b. 10
 - c. 15
 - d. 18
- The blow off pressure of a safety valve is
 - a. slightly more than atmospheric pressure
 - b. equal to boiler working pressure
 - c. half of the boiler working pressure
 - d. 106% of the boiler working pressure
- A type of firing in pulverized coal-fired furnaces which employs a turbulent burner, the unit being located in the front or near wall.
 - a. vertical firing
 - b. horizontal firing
 - c. impact firing
 - d. tangential firing
- Is the difference between saturation temperature and actual temperature
 - a. Degree subcooling
 - b. Degree superheating
 - c. Compressed temperature
 - d. Saturation temperature
- The actual horsepower delivered by the engine to the drive shaft -
 - a. indicated horsepower
 - b. brake horsepower
 - c. friction horsepower
 - d. none of the above
- Peak load for a period of time divided by installed capacity is:
 - a. capacity factor
 - b. demand factor
 - c. utilization factor
 - d. load factor
- The size of a steam reciprocating pump is generally designated by three-digit number such as 646. What would be the first number designate?
 - a. stroke of the pump in inches
 - b. inside diameter of the steam cylinder measured in inches
 - c. percent clearance
 - d. number of cylinders
- Fluids that are pumped in processing work are frequently more viscous than water. Which of the following statement is correct?
 - a. Reynolds number varied directly as viscosity
 - b. Efficiency of a pump increases as the viscosity increases
 - c. Increased fluid friction between the pump parts and the passing fluid increases useful work
 - d. Working head increases as the viscosity increases
- The velocity at which the flow of a liquid reversibly changes from viscous to turbulent.
 - a. low velocity
 - b. high velocity
 - c. velocity head
 - d. critical velocity
- The portion of sunlight, rich in ultraviolet rays, which has a strong effect on the photographic plate.
 - a. gamma rays
 - b. beta rays
 - c. X-rays
 - d. Actinic rays

- Which of the following is more accurate about streamline?
 - a. It is a path of fluid particle.
 - b. It is a line normal to the velocity vector everywhere
 - c. It is fixed in space in steady flow
 - d. It is perpendicular to the path line
- The function of the sharp seal in an open type refrigerating system is to prevent the gas from escaping the:
 - a. evaporator
 - b. Compressor
 - c. Condenser
 - d. Capillary tube
- When the liquid boils at a pressure below atmospheric, the liquid has reached its:
 - a. boiling point
 - b. pressure point
 - c. critical point
 - d. Vapor pressure
- The cylinder head and the cylinder block is fitted to a water and gas tight seal through -
 - a. the cylinder head's bolts and nuts
 - b. fitting the cylinder head and block
 - c. cylinder head gasket
 - d. none of the above
- Another term for sludge formation -
 - a. carbon residue
 - b. oxidation
 - c. water and sediments
 - d. pour point
- The hollowed spaces within the cylinder head directly above the piston where the fuel change is burned
 - a. piston head
 - b. top dead center
 - c. combustion chamber
 - d. valve head
- The power output of the engine is increased through -
 - a. scavenging
 - b. turbo-charging
 - c. supercharging
 - d. all of the above
- The property of lubricating oil which indicates its relative fluidity -
 - a. viscosity
 - b. no gumming
 - c. waterless and free from sediments
 - d. verification of contents
- Factors which affects the efficiency of diesel power in stationary plant -
 - a. financial allocation
 - b. space allotment
 - c. supply of power
 - d. none of the above
- The cooling system in which water under pressure flows through the engine jackets and is drained off or recirculated through a cooling tower -
 - a. open cooling system
 - b. closed cooling system
 - c. open and closed cooling system
 - d. all of the above
- What kind of heat exchanger where water is heated to a point that dissolved gases are liberated?
 - a. evaporator
 - b. condenser
 - c. intercooler
 - d. deaerator
- An air current in a confined space such as that in a cooling tower or chimney is called:
 - a. variable flow
 - b. velocity profile
 - c. velocity
 - d. draft
- One of the following tasks which is an example of preventive maintenance is:
 - a. cleaning up the cup on a rotary cup burner
 - b. cleaning a completely clogged oil strainer
 - c. replacing a leaking valve
 - d. replacing a blown fuse
- What should be the temperature of both the water and steam whenever they are present together?
 - a. saturation temperature for the existing pressure
 - b. boiling point of water at 101.325 Kpa
 - c. super heated temperature
 - d. one hundred degree centigrade
- How many pounds of air are theoretically needed to burn one pound of diesel fuel oil?
 - a. 28
 - b. 14
 - c. 18
 - d. 22
- An unloader is used on air compressor to:
 - a. relief air pressure
 - b. start easier
 - c. stop easier
 - d. run faster
- The relationship of water vapor in the air at the dew point temperature to the amount that should be in the air if the air were saturated at the dry-bulb temperature is:
 - a. partial pressure actual dew point
 - b. percentage humidity
 - c. relative humidity
 - d. run faster
- The brake effective pressure of an internal combustion engine is:
 - a. obtained from the indicator card
 - b. greater than the indicated mean effective pressure
 - c. a computed value from brake measurements
 - d. can be determined without knowing the engine speed

- A chemical action which causes destruction of the surface of metal by oxidation.
 - a. emulsification
 - b. scaling
 - c. formation
 - d. corrosion
- A device that changes electric current from high to low voltage
 - a. alternator
 - b. generator
 - c. transformer
 - d. dynamometer
- An automatic device which allows the passage of water but prevents the passage of steam.
 - a. steam trap
 - b. fusible plug
 - c. separator
 - d. blow-off
- A chart which compares visually smoke density with graded color bars.
 - a. Mollier's chart
 - b. Achumann chart
 - c. control chart
 - d. Ringleman chart
- In the hydel plant classified as medium head pressure due to water backflow may be delivered by:
 - a. forebay
 - b. Wicket gate
 - c. surge tank
 - d. draft tube
- The proportions of carbon, hydrogen, oxygen, nitrogen, sulfur and ash is known as:
 - a. Proximate analysis
 - b. Orsat Analysis
 - c. Ultimate analysis
 - d. Molal Analysis
- The ratio of the dynamic depression head to the total head of a pump.
 - a. Heat factor
 - b. Cavitation factor
 - c. Discharge coefficient
 - d. Diagram factor
- A steam engine that has two or more cylinders, the exhaust steam from one cylinder entering the next cylinder at a lower pressure is known:
 - a. compound engine
 - b. wankel engine
 - c. reciprocating engine
 - d. otto engine
- $(U + PV)$ is a quantity called:
 - a. flow energy
 - b. shaft work
 - c. entropy
 - d. enthalpy
- Work of energy can be a function of all of the following, EXCEPT:
 - a. temperature and entropy
 - b. torque and angular rotation
 - c. force and time
 - d. power and time
- Which of the following thermodynamic cycle is the most efficient?
 - a. Carnot
 - b. Brayton
 - c. Otto
 - d. Diesel
- Enthalpy in an ideal gas is a function only of:
 - a. temperature
 - b. entropy
 - c. Internal energy
 - d. pressure
- From the standpoint of thermodynamics, the best compressor follows which compressor curve:
 - a. $PV = C$
 - b. $PV^k = C$
 - c. $PV^{1.3} = C$
 - d. $PV^{1.25} = C$
- Temperature above which a given gas cannot be qualified:
 - a. Absolute temperature
 - b. Cryogenic temperature
 - c. Critical temperature
 - d. Boiling temperature
- The possibility of detonation in a spark ignition will be increased by:
 - a. Lowering compression ratio
 - b. Lowering the inlet air temperature
 - c. Lowering the jacket water temperature
 - d. advancing the spark time
- Which of the following is a natural draft cooling tower.
 - a. Cooling pond
 - b. Induced draft cooling tower
 - c. evaporative cooling tower
 - d. Hyperbolic cooling tower
- Cavitation is the result of:
 - a. Static pressure in a fluid becoming less than fluid vapor pressure
 - b. Rivets under impact load
 - c. Exposure of concrete to salt water
 - d. Heat treatment of a low carbon steel
- Which of the metal is the best heat conductors?
 - a. aluminum
 - b. copper
 - c. gold
 - d. silver
- When exposed to the atmosphere, which of the following liquids is the coldest?
 - a. oxygen
 - b. argon
 - c. hydrogen
 - d. helium
- The voltage of a galvanic cell does not depend on which of the following parameters?
 - a. concentration of solutions
 - b. temperature
 - c. pressure

- d. volume
- Brayton cycle cannot be used in reciprocating engine oven for the same compression ratio and work output because:
 - a. Brayton cycle is more efficient
 - b. Otto cycle is less efficient
 - c. Brayton cycle is for slow speed engines only
 - d. Large volume of low pressure gas cannot be efficiently handled in reciprocating engines
- Total sulfur content in a diesel fuel oil must not exceed:
 - a. 0.1%
 - b. 0.5%
 - c. 0.2%
 - d. 0.15%
- The mean effective pressure of a diesel cycle having fixed compression ratio will increase if cut-off ratio is:
 - a. Increased
 - b. decreased
 - c. Independent of CR
 - d. depends on other factors
- Boiling point of Freon 22 is:
 - a. -33.33°C
 - b. -29.8°C
 - c. -78.5°C
 - d. -40°C
- Which refrigerant has the highest critical point temperature?
 - a. Ammonia
 - b. Freon 11
 - c. Freon 12
 - d. Freon 22
- A material designed to reduce friction between surfaces.
 - a. Shims
 - b. Lubricant
 - c. Fuel
 - d. Bearing
- An instrument designed to measure the rotational speed of a machine.
 - a. Tachometer
 - b. Governor
 - c. Rotator
 - d. Anemometer
- When gas is heated at constant volume, the process is called:
 - a. isotropic
 - b. isobaric
 - c. isometric
 - d. adiabatic
- An engine cycle containing two adiabatic and two isothermal preocesses.
 - a. Carnot cycle
 - b. Rankine cycle
 - c. Diesel cycle
 - d. Otto cycle
- $C_p = C_v + R$ applies to:
 - a. ideal gas
 - b. two-phase states
 - c. enthalpy
 - d. all pure substance
- When droplets of water are carried by steam in the boiler
 - a. priming
 - b. foaming
 - c. carryover
 - d. embrittlement
- The chemical formula for methyl chloride
 - a. $\text{C}_2\text{H}_5\text{Cl}$
 - b. CH_3Cl
 - c. CH_2Cl_2
 - d. CHCl_2
- One type of pipe which is not recommended for use in ammonia refrigerating system is:
 - a. steel
 - b. brass
 - c. seamless
 - d. iron
- A device that changes electric current from high to low voltage
 - a. carburetor
 - b. separator
 - c. transformer
 - d. regulator
- The part of the boiler design for burning the fuel
 - a. oven
 - b. heater
 - c. furnace
 - d. gage cock
- The first law of thermodynamics state that:
 - a. Heat energy cannot be completely transformed into work
 - b. Internal energy is due to molecular motion
 - c. Energy can neither be created nor destroyed
 - d. Entropy of the universe is increased by irreversible process
- Clearing a cylinder of exhaust gases by forcing into it a current of air switch provides clean air for the next compression stroke of an engine.
 - a. turbo charging
 - b. purging
 - c. scavenging
 - d. emulsifying
- Coals that become soft under the usual furnace temperature and merge into undesirable masses of coke
 - a. anthracite
 - b. bituminous
 - c. lignite
 - d. caking

- A mechanical device used in refrigeration system for the purpose of increasing the pressure upon the refrigerant.
 - a. evaporator
 - b. condenser
 - c. expansion coil
 - d. compressor
- In power piping system, the corresponding piping color for proper fluid identification for steam is:
 - a. green
 - b. silver gray
 - c. light blue
 - d. violet
- The name given to procedure by which efficiency of a combustion process is determined.
 - a. chamber
 - b. heat balance
 - c. oxidation
 - d. evolution
- The evaporator constructed of pipe or tubing
 - a. fusible plug
 - b. steam trap
 - c. expansion coil
 - d. sling
- A device having a pre-determined temperature fusible member for the relief of pressure
 - a. fusible plug
 - b. stop valve
 - c. water column
 - d. pressure valve
- The process of supplying or removing air by natural or mechanical means to or from any surface
 - a. refrigeration
 - b. humidification
 - c. circulation
 - d. ventilation
- The maximum load carried by a system or a unit of equipment over a designated period of time
 - a. average load
 - b. peak load
 - c. nominal rating
 - d. plant
- As commonly used, the PH value:
 - a. has no relation to the hydrogen - ion concentration
 - b. for boiler water is usually kept at a value between 5 and 6
 - c. will be lower if boiler is treated with calistic soda
 - d. of 7 represent a neutral solution
- Which of the following is not a viscosity rating?
 - a. Centistroke
 - b. Redwood
 - c. SSU
 - d. Degree API
- Term not used in relation to lubrication oil:
 - a. Cetane number
 - b. Conradson carbon
 - c. Neutralization number
 - d. Demulsibility
- It is the temperature at which vapor forms above the liquid fuel:
 - a. Diesel index
 - b. Fire point
 - c. Flash point
 - d. Critical point
- For which type of process does the equation $dQ = TDS$ hold?
 - a. Irreversible
 - b. Isothermal
 - c. reversible
 - d. isobaric
- During an adiabatic, internally reversible process, what is true about the change of entropy?
 - a. it is always zero
 - b. it is always less than zero
 - c. it is always greater than zero
 - d. it is temperature - dependent
- A boiler furnishing hot water to be used externally to itself.
 - a. water heating
 - b. hot water supply boiler
 - c. low pressure heating boiler
 - d. portable boiler
- A refrigerating system in which the evaporator is in direct contact with the refrigerated material or space or is located.
 - a. direct expansion
 - b. chilled water
 - c. flooded
 - d. multiple
- Trichloromonofluoromethane is also known as:
 - a. Freon 11
 - b. Freon 12
 - c. Freon 21
 - d. Freon 114
- Dichlorodifluoromethane is otherwise known as:
 - a. Freon 11
 - b. Freon 12
 - c. Freon 21
 - d. Freon 114
- An internally fired boiler which is self-contained and primarily intended for temporary location.
 - a. low pressure boiler
 - b. unified pressure boiler
 - c. portable boiler
 - d. power boiler
- A closed vessel in which pressure is obtained from an indirect application of heat.
 - a. low pressure boiler
 - b. unified pressure boiler
 - c. portable boiler
 - d. power boiler

- A refrigerating system using direct method in which refrigerant is delivered to two or more evaporators in separate rooms or refrigerators
 - a. direct expansion
 - b. chilled water
 - c. flooded
 - d. multiple
- The ratio of the energy removed at the evaporator to the energy supplied to the compressor
 - a. TOR
 - b. Power ratio
 - c. Compression ratio
 - d. COP
- Dichloromethane is known as:
 - a. Methyl Chloride
 - b. Methyl formula
 - c. Methylene chloride
 - d. Ethyl Chloride
- The alcohol most frequently considered as fuel for internal combustion engines
 - a. Methanol
 - b. Grain alcohol
 - c. isopropanol
 - d. isoprophyl alcohol
- The temperature at which the fuel gives off vapors ignited by an open flame.
 - a. pour point
 - b. power point
 - c. flash point
 - d. boiling point
- In the vapor-compression refrigeration cycle, the refrigerant leaving the expansion valve is:
 - a. liquid
 - b. partly vaporized
 - c. saturated vapor
 - d. superheated vapor
- The capacity of an air compressor is:
 - a. Independent on the rpm of the compressor
 - b. decrease with lower discharge pressure
 - c. may be controlled by varying the clearance value
 - d. increases with lower values of the polytropic constant
- Mol ratio is proportional to the volumetric ratio is known as:
 - a. Amogat's Law
 - b. Pascal's Law
 - c. Avogadro's Law
 - d. Archimedes' Law
- Heat movement through space by shining, just as light is radiated.
 - a. Convection
 - b. Circulation
 - c. Conduction
 - d. Radiation
- The primary object of the washing of coal is to decrease
 - a. dustiness
 - b. ash fusion temperature
 - c. ash and sulphur
 - d. volatile matter
- A periodic, time-scheduled inspections to detect wear and tear in advance of actual equipment breakdown with subsequent replacement of worn-out parts before they fail in order to minimize interference in operation
 - a. routine maintenance
 - b. preventive maintenance
 - c. corrective maintenance
 - d. general maintenance
- A part of the generating capacity of a hydro-electric plant that is connected to the bus and ready to take load.
 - a. cold reserve
 - b. hot reserve
 - c. standing reserve
 - d. spinning reserve
- Which of the following statements about Mach number (M) is false?
 - a. Mach number is the ratio of velocity to the sonic velocity
 - b. Mach 11 is the maximum attainable velocity in a nozzle throat
 - c. Supersonic velocities ($M > 1$) are achievable in the diffuser section of a rocket nozzle if expansion ratio is great enough
 - d. Mach no. has the dimension of velocity
- The hydraulic formula $CA\sqrt{2gh}$ is used to find the:
 - a. quantity of discharge through an orifice
 - b. length of pipe in a closed network
 - c. velocity of flow in a co=losed conduit
 - d. friction factor of a pipe
- If gas is expanded isothermally in a cylinder
 - a. No work is done
 - b. Heat must have been abstracted to make the process possible
 - c. Heat must have been added during the process
 - d. It must have been completely insulated
- All of the following statements about wet bulb temperature are true, EXCEPT
 - a. Wet bulb temperature equals adiabatic saturation temperature
 - b. Wet bulb temperature is the only temperature necessary to determine grains water per lb of dry air
 - c. Wet bulb temperature lies numerically between dew point and dry bulb temperature for unsaturated systems
 - d. Wet bulb temperature equals both dry-bulb and dew point temperature at 100% relative humidity
- In a four-cycle gasoline engine, when properly tuned, the valve operation will be such that the:
 - a. Intake-opens after the dead center
 - b. Exhaust opens before bottom dead center

- c. Intake closes before bottom dead center
 - d. Exhaust closes before bottom dead center
- Which of the following is member of the halogen family?
 - a. sodium
 - b. fluorine
 - c. hydrogen chloride
 - d. phosphorus
- A sample substance to which a chemical formula can be assigned, whose weight is equal to its formula weight is formed a:
 - a. molecule
 - b. mole
 - c. grain equivalent weight
 - d. atom
- Hydrogen is common to all:
 - a. acids
 - b. salts
 - c. oxides
 - d. metals
- Which of the following does not illustrate the effect of temperature or pressure on gas solubility?
 - a. Air bubbles form on the sides of a warm glass of water
 - b. Soda pop is bottled under pressure
 - c. Boiling frees water gases
 - d. Air is more humid on rainy days
- The chemical process which occurs when water is added to cement is
 - a. oxidation
 - b. Brownian movement
 - c. plastic flow
 - d. hydration
- To determine the chemical formula of a compound, we need to know all of the following EXCEPT the:
 - a. elements that compose the compound
 - b. atomic weights of the elements in a compound
 - c. molecular weight of the compound
 - d. density of the compound
- From the 1993 PSME Code, electricity is represented by the color:
 - a. light blue
 - b. Violet
 - c. White
 - d. Light orange
- For the steam jet refrigeration system, the refrigerant used is:
 - a. Freon 11
 - b. Freon 12
 - c. Refrigerant 718
 - d. None of these
- Heat added to or removed from a substance that causes a change of temperature:
 - a. absolute heat
 - b. specific heat
 - c. latent heat
 - d. sensible heat
- The ratio of the maximum load to the plant capacity is:
 - a. plant capacity factor
 - b. use factor
 - c. utilization factor
 - d. demand factor
- Orsat analysis is:
 - a. volumetric analysis of the wet gas
 - b. dry exhaust gas volumetric analysis
 - c. ultimate analysis of fuel
 - d. molal analysis of fuel
- If the rpm of the pump is doubled, the power required is:
 - a. quadrupled
 - b. doubled
 - c. remains the same
 - d. increases exponentially
- A closed vessel intended for application of heat to generate steam or other vapor to be used externally to itself
 - a. boiler
 - b. furnace
 - c. oven
 - d. heater
- Mixture of hydrocarbon liquefied under pressure for efficient transportation, storage, and use
 - a. gasoline
 - b. LPG
 - c. TIG
 - d. Tar
- A valve held closed by a spring or other means and designed to automatically relieve pressure in excess of its setting
 - a. stop valve
 - b. pressure relief device
 - c. spring valve
 - d. stop cock
- Transfer of heat between a fluid and a surface by the circulation or mixing of the fluid.
 - a. Convection
 - b. Conduction
 - c. Friction
 - d. Fusion
- A substance that absorbs heat through expansion or vaporization
 - a. lubricant
 - b. relaxant
 - c. refrigerant
 - d. fuel
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DEOPHANTINE (K)